Couse Code: CSS3913 Computer Vision and Image Processing

Topic 1: Introduction to Computer Vision

ABDULWAHAB FUNSHO ATANDA

About the course

Objective:

To provide the students with advanced knowledge and techniques of computer vision that simulates the biological vision, so that the students are able to develop software that is used in computer vision field.

Course Learning Outcomes:

- Explain methods for decomposing an image into basic elements: edges, regions, various higher order image features.
- Apply the techniques for extracting information from digital imagery.
- Build application by using computer vision algorithm.

About the course

Topic Covers:

- Introduction to Computer Vision
- Digital Image
- Image Enhancement
- Convolution
- Labelling
- Grouping
- Matching
- Transforming Image Representations
- Advance Techniques
- Introduction to Pattern Recognition

Main Reference

Gonzalez and Woods. Digital Image Processing, 4th Edition, Pearson, 2018.

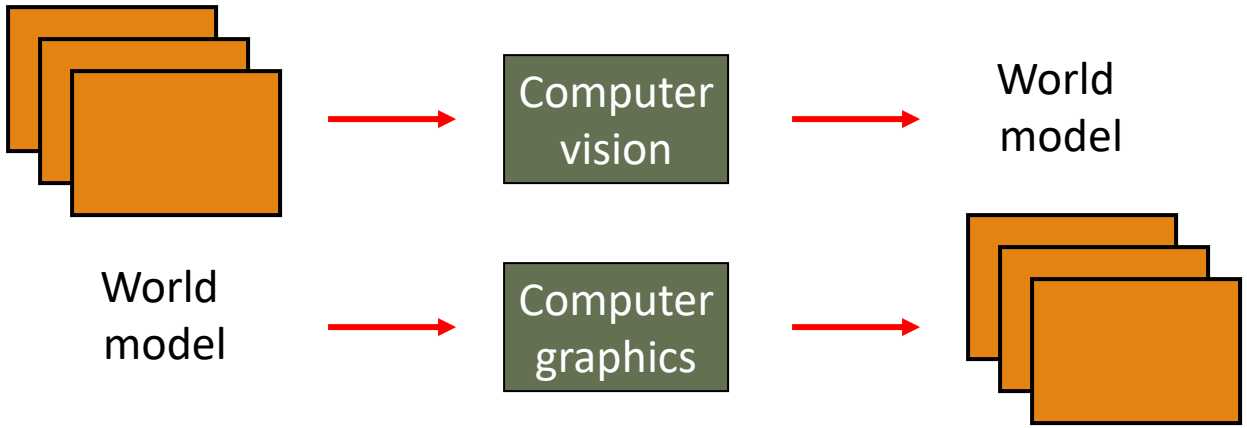
Parker, J. & Terzidis, K. (2011). Algorithms for image processing and computer vision. Indianapolis, Ind: Wiley Pub. ISBN: 978-0-470-64385-3.

Additional references supporting the course:

Petrou, M. & Petrou, C. (2010). Image processing : the fundamentals. Chichester, U.K: Wiley. ISBN: 978-0-470-74586-1.

What is Computer Vision?

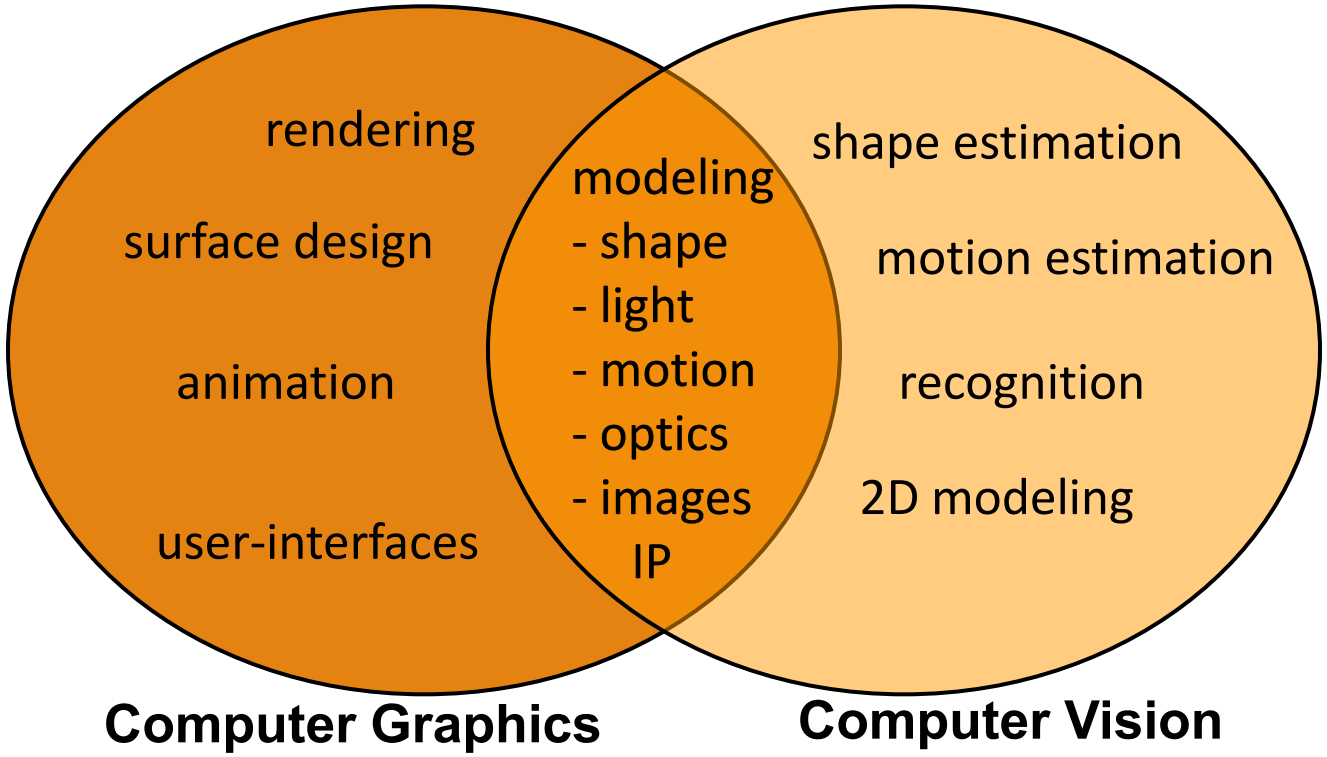
- Image Understanding (AI, behavior)
- A sensor modality for robotics
- Computer emulation of human vision
- Inverse of Computer Graphics



What is Computer Vision?

- **Image Understanding (AI, behavior)**
- Computer vision is seen as a branch of AI focused on *understanding images*.
- For interpreting what is happening in a scene, recognizing objects, actions, and even inferring behaviors.
- Example: detecting pedestrians and vehicles in autonomous driving to predict behavior.
- **A Sensor Modality for Robotics**
- Here, vision is treated as another *sensor* for robots, similar to LIDAR, sonar, or touch.
- Robots can “see” and use images to navigate, manipulate objects, and interact with their environment.
- Example: robotic arms using cameras to pick up irregularly shaped items.
- **Computer Emulation of Human Vision**
- This perspective emphasizes *mimicking biological vision*.
- The goal is to replicate how humans perceive depth, motion, color, and patterns.
- Example: facial recognition systems inspired by how humans identify people.
- **Inverse of Computer Graphics**
- Computer graphics takes a model (3D shapes, textures, light) and produces an image.
- Computer vision works in the opposite direction: starting with an image and reconstructing the structure, objects, or 3D world behind it.
- Example: 3D reconstruction from 2D images in medical imaging or augmented reality.

Intersection of Vision and Graphics

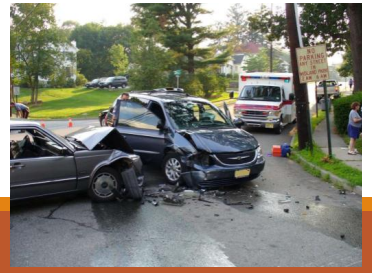


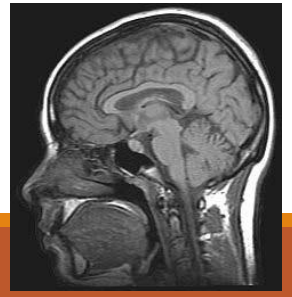
What is Computer Vision?

Computer vision is the science and technology of machines that see.

Concerned with the theory for building artificial systems that obtain information from images.

The image data can take many forms, such as a video sequence, depth images, views from multiple cameras, or multi-dimensional data from a medical scanner





Computer Vision vs Image Processing

Aspect	Computer Vision	Image Processing
Goal	Extract meaning and understanding from images (e.g., recognize objects, actions, 3D structures).	Enhance, transform, or analyze images at a pixel level without necessarily understanding content.
Focus	High-level interpretation – answering "What is in the image?"	Low-level operations – answering "How can we improve or modify the image?"
Examples of Tasks	Object detection, face recognition, scene understanding, 3D reconstruction.	Noise removal, contrast enhancement, edge detection, image compression.
Techniques	Machine learning, deep learning, pattern recognition, geometry-based inference.	Filtering, Fourier transforms, histogram equalization, morphological operations.
Input/Output	Input: image(s) → Output: semantic information (labels, 3D models, decisions).	Input: image(s) → Output: processed image(s) (improved or modified).
Relation to AI	Strongly tied to Artificial Intelligence and robotics (enables decision-making).	Not necessarily AI-related; mainly mathematical and algorithmic.
Complexity	Generally higher complexity (requires models, training, inference).	Comparatively simpler (rule-based, deterministic).
Example Applications	Self-driving cars, medical diagnosis (tumor detection), augmented reality.	Medical image enhancement, satellite image sharpening, photo editing.

Computer Vision

Make computers understand images and videos.

What kind of scene?

Where are the cars?

How far is the building?

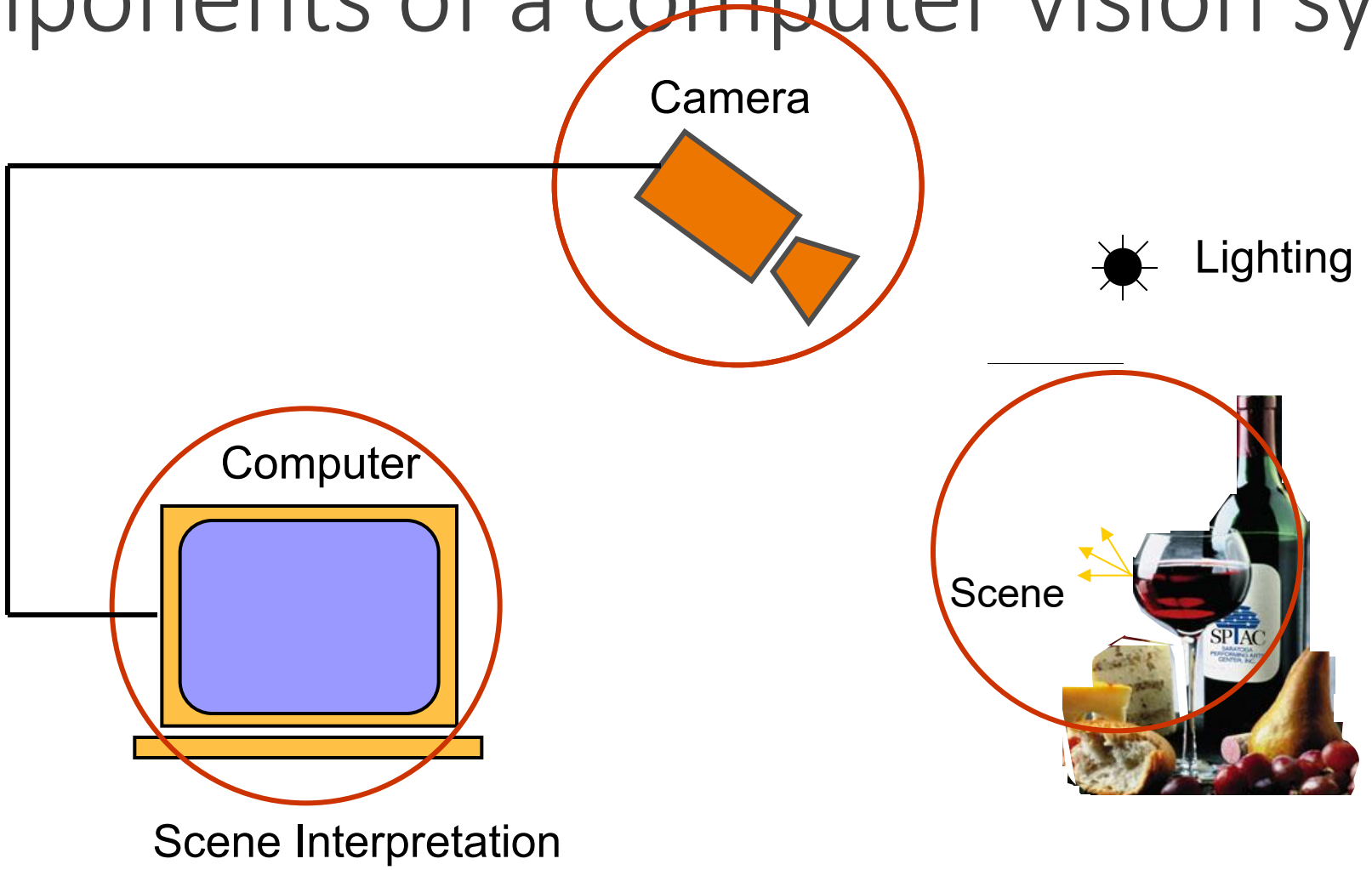
…

Computer Vision vs Machine Vision

Machine Vision is the application of computer vision to industry, manufacturing, surveillance and etc.(growing number of applications).

Whereas **Computer Vision** is mainly focused on machine-based image processing, machine vision most often requires also digital input/output devices and computer networks to control other manufacturing equipment such as robotic arms. Machine Vision is a subfield of engineering that encompasses computer science, optics, mechanical engineering, and industrial automation.

Components of a computer vision system



Computer vision vs human vision

What we see

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

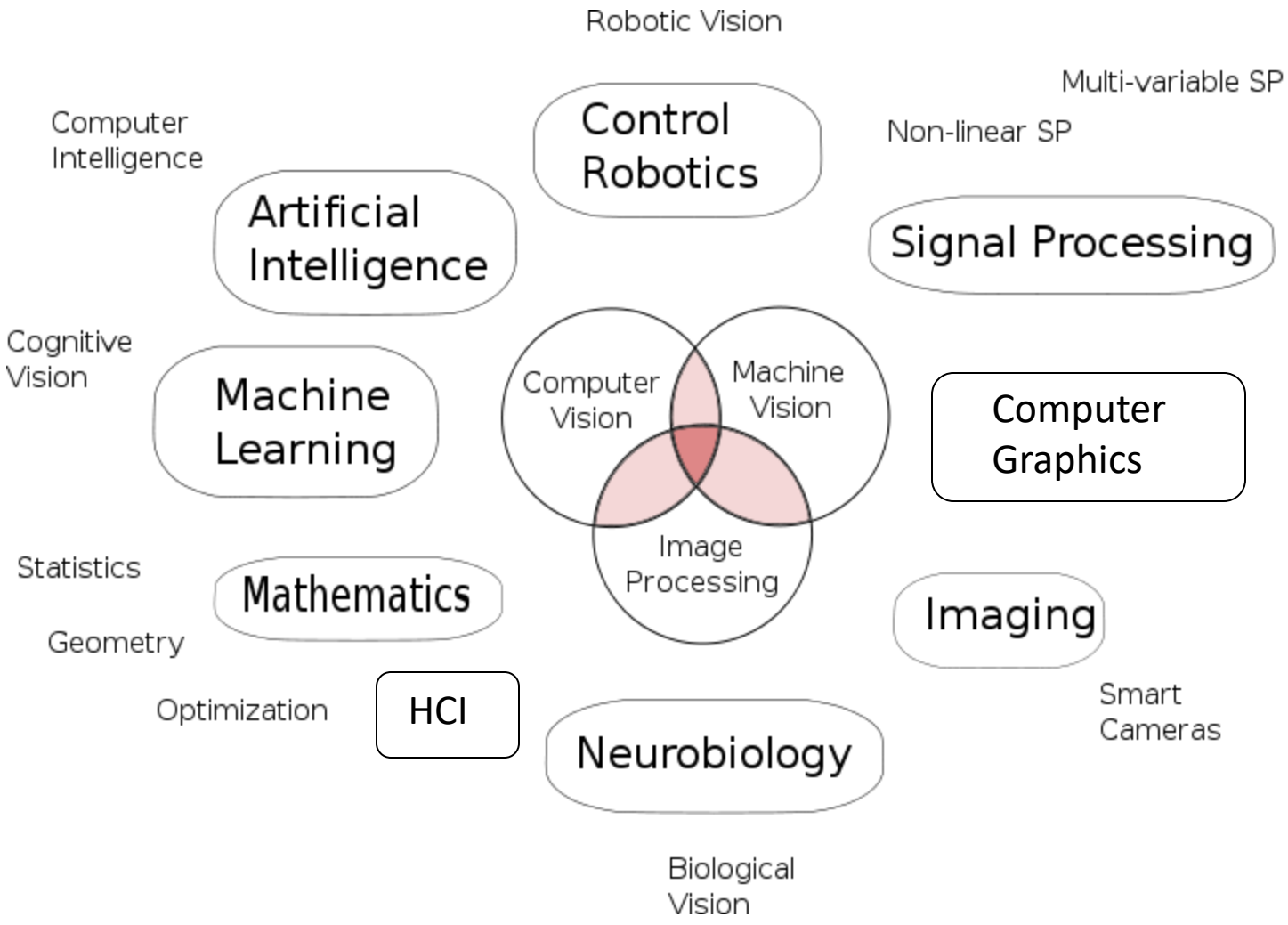
What a computer sees

Vision is really hard

Vision is an amazing feat of natural intelligence

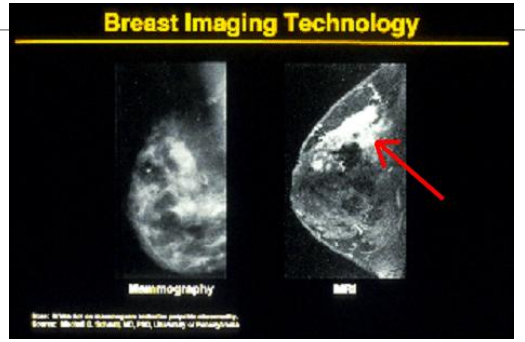
- Visual cortex occupies about 50% of Macaque brain
- More human brain devoted to vision than anything else

Vision is multidisciplinary



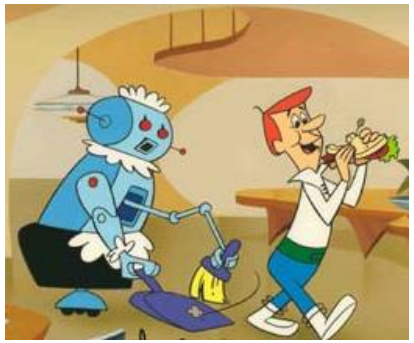
Why computer vision matters

Safety



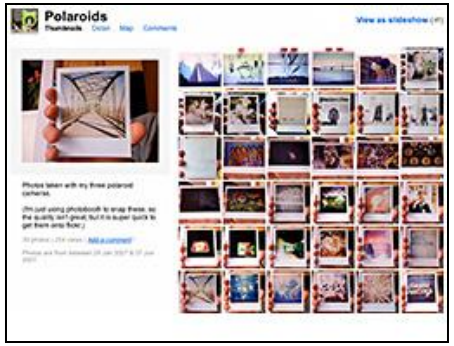
Health

Security



Comfort

Fun



Access

A little story about Computer Vision

In 1966, Marvin Minsky at MIT asked his undergraduate student Gerald Jay Sussman to “spend the summer linking a camera to a computer and getting the computer to describe what it saw”. We now know that the problem is slightly more difficult than that. (Szeliski 2009, Computer Vision)

Ridiculously brief history of computer vision

1966: Minsky assigns computer vision as an undergrad summer project

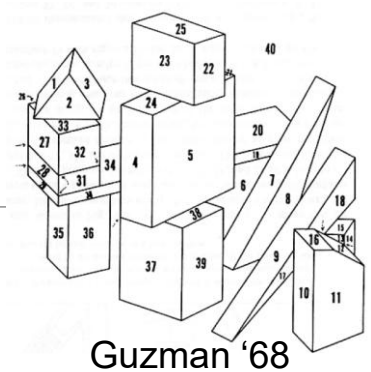
1960's: interpretation of synthetic worlds

1970's: some progress on interpreting selected images

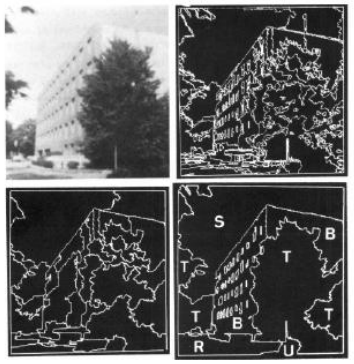
1980's: ANNs come and go; shift toward geometry and increased mathematical rigor

1990's: face recognition; statistical analysis in vogue

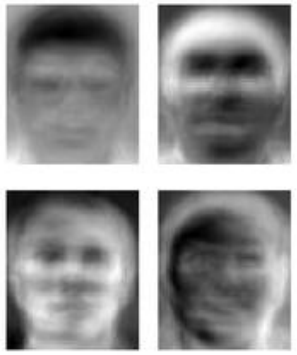
2000's: broader recognition; large annotated datasets available; video processing starts; vision & graphis; vision for HCI; internet vision, etc.



Guzman '68



Ohta Kanade '78

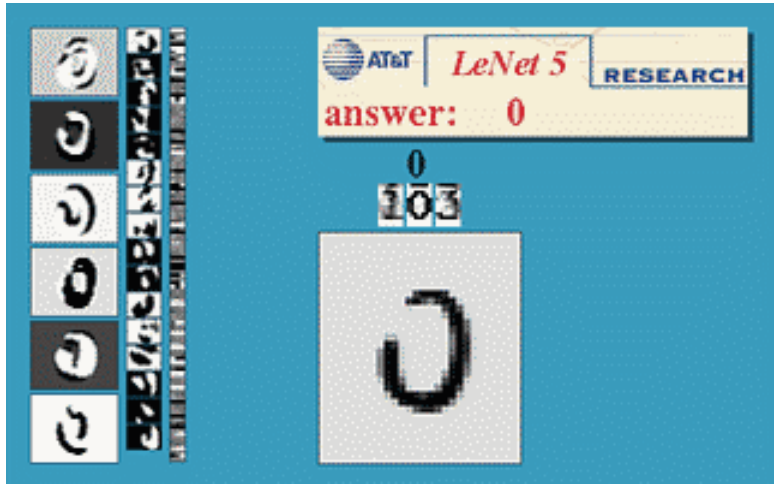


Turk and Pentland '91

Optical character recognition (OCR)

Technology to convert scanned docs to text

- If you have a scanner, it probably came with OCR software



Digit recognition, AT&T labs
http://www.research.att.com/~yann/

License plate readers
http://en.wikipedia.org/wiki/Automatic_number_plate_recognition

Face detection

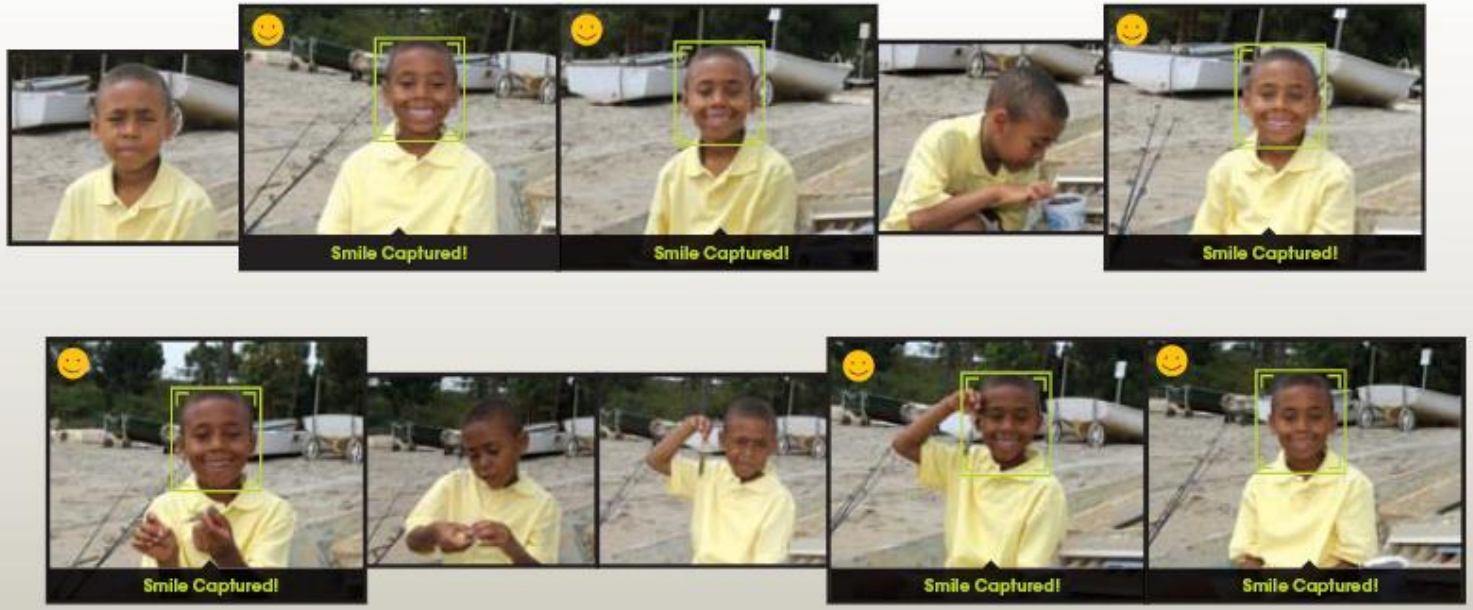
Many new digital cameras now detect faces

- Canon, Sony, Fuji, …

Smile detection

The Smile Shutter flow

Imagine a camera smart enough to catch every smile! In Smile Shutter Mode, your Cyber-shot® camera can automatically trip the shutter at just the right instant to catch the perfect expression.



Sony Cyber-shot® T70 Digital Still Camera

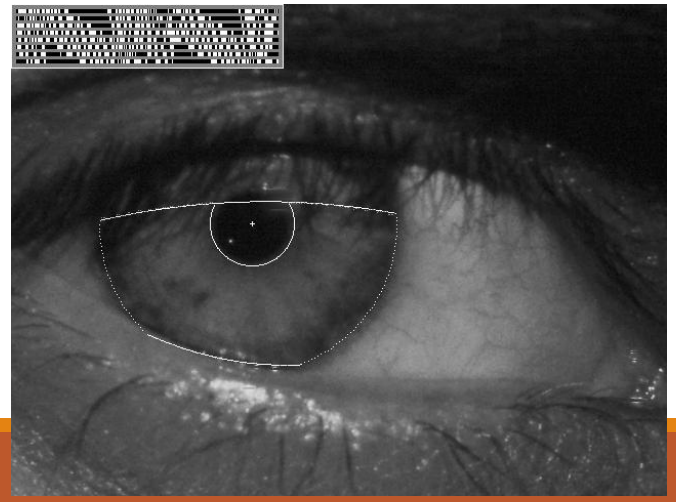
Object recognition (in supermarkets)

LaneHawk by EvolutionRobotics

“A smart camera is flush-mounted in the checkout lane, continuously watching for items. When an item is detected and recognized, the cashier verifies the quantity of items that were found under the basket, and continues to close the transaction. The item can remain under the basket, and with LaneHawk,you are assured to get paid for it… “

Vision-based biometrics

"*How the Afghan Girl was Identified by Her Iris Patterns*" Read the story wikipedia



Login without a password…

Fingerprint scanners on many new laptops, other devices

Face recognition systems now beginning to appear more widely
http://www.sensiblevision.com/

Special effects: shape capture

The Matrix movies, ESC Entertainment, XYZRGB, NRC

Special effects: motion capture

Pirates of the Carribean, Industrial Light and Magic

Sports

Sportvision first down line
Nice explanation on www.howstuffworks.com

http://www.sportvision.com/video.html

Smart cars

Slide content courtesy of Amnon Shashua

Mobileye [wiki article]

- Vision systems currently in high-end BMW, GM, Volvo models
- By 2010: 70% of car manufacturers.

Google cars

http://www.nytimes.com/2010/10/10/science/10google.html?ref=artificialintelligence

Vision in space

NASA'S Mars Exploration Rover Spirit captured this westward view from atop a low plateau where Spirit spent the closing months of 2007.

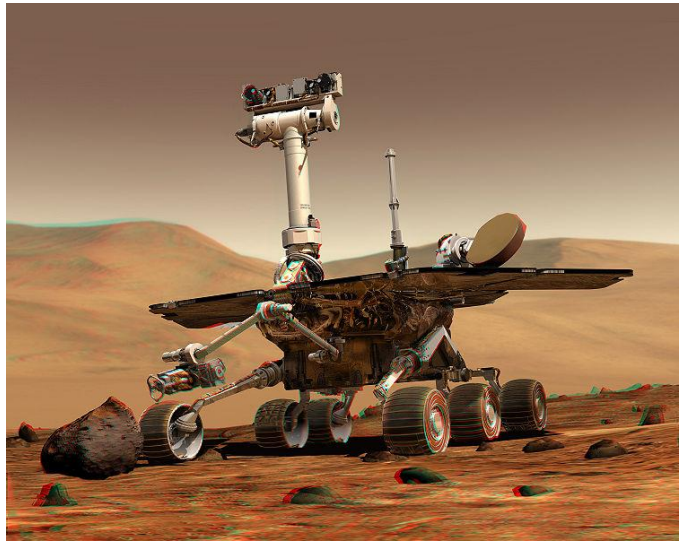
Vision systems (JPL) used for several tasks

- Panorama stitching
- 3D terrain modeling
- Obstacle detection, position tracking
- For more, read "Computer Vision on Mars" by Matthies et al.

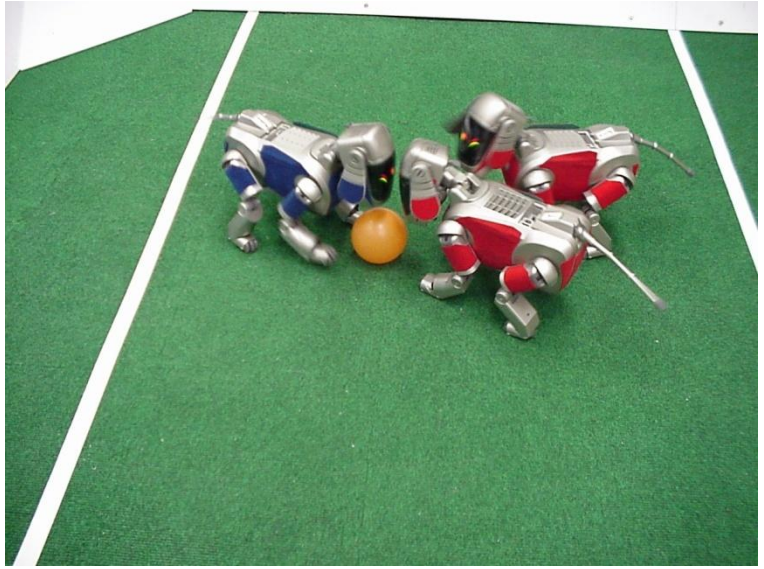
Industrial robots

Vision-guided robots position nut runners on wheels

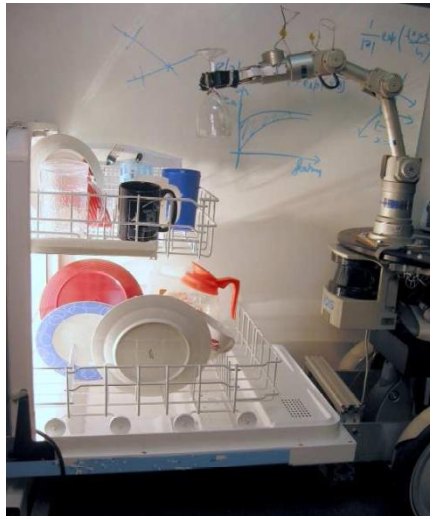
Mobile robots



NASA's Mars Spirit Rover
http://en.wikipedia.org/wiki/Spirit_rover

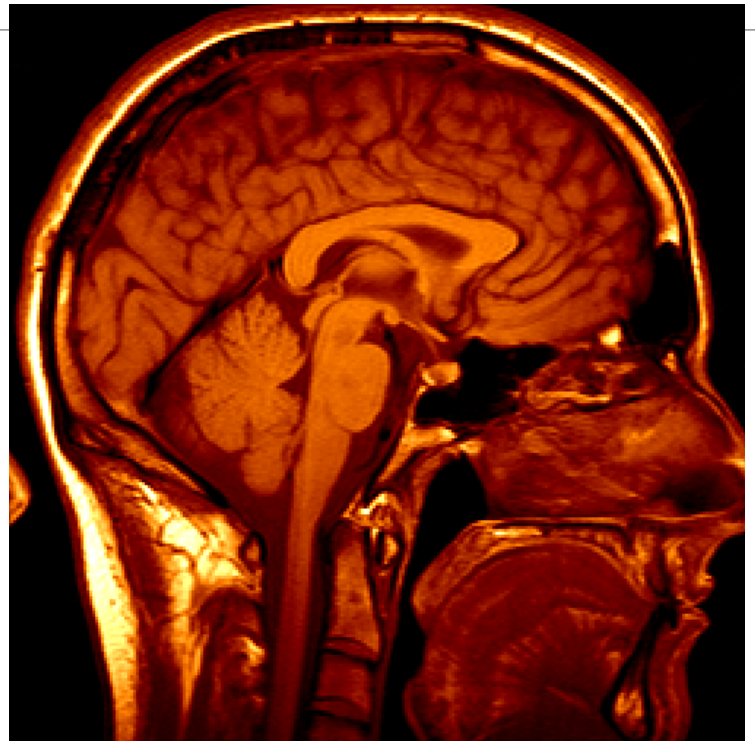


http://www.robocup.org/



Saxena et al. 2008
STAIR at Stanford

Medical imaging



3D imaging
MRI, CT

Image guided surgery
Grimson et al., MIT

Computer Vision Processes

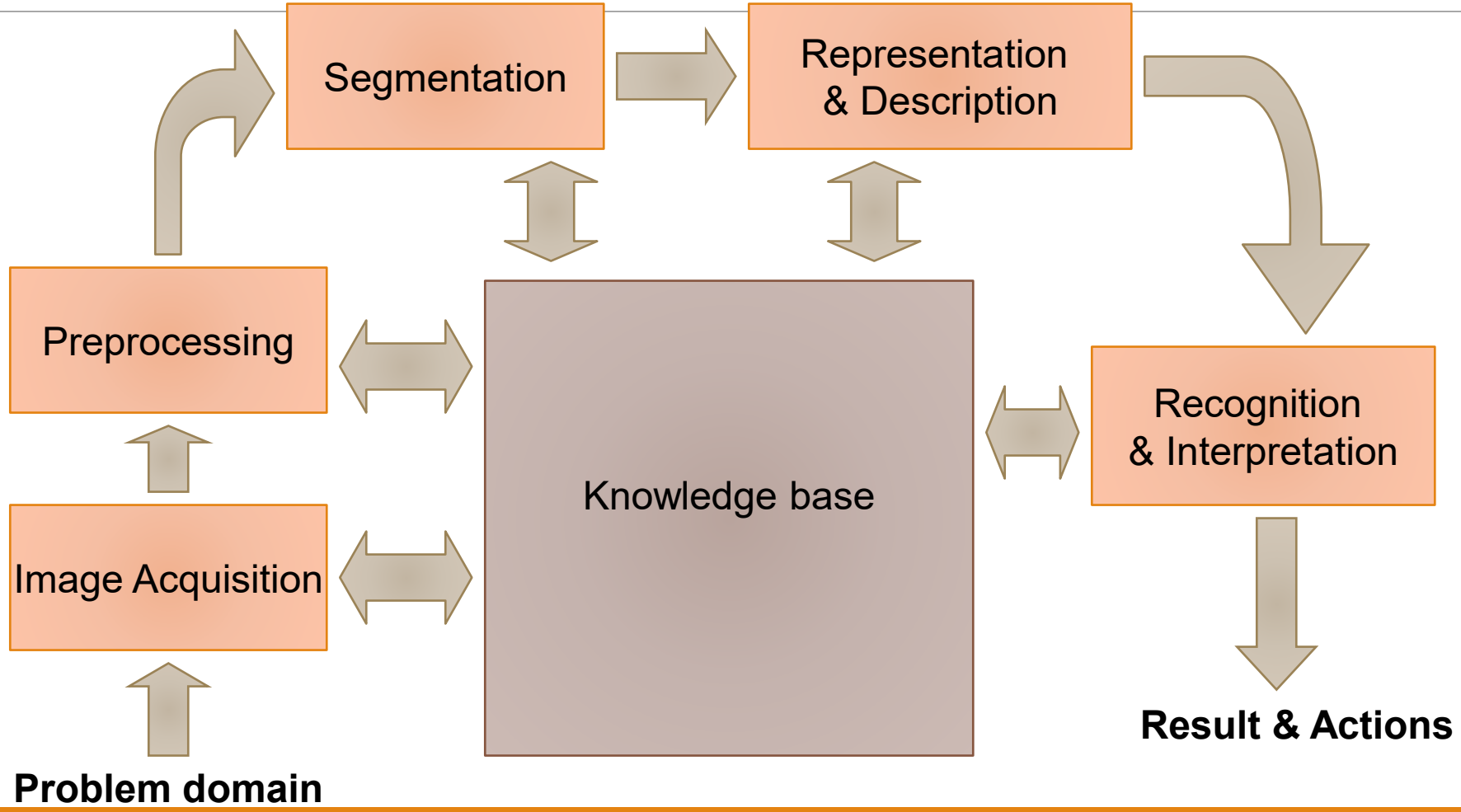


Image Acquisition

- The 1st step in the CV process → ***Image Acquisition*** (acquire a digital image).
- Requires an imaging sensor and the capability to digitize the signal produced by the sensor.
- The sensor could be a monochrome or color camera that produces an entire image of the problem domain.
- If the output of the camera or other imaging sensor is not already in digital form, an analog-to-digital converter digitizes it.
- The nature of the sensor and the image it produces are determined by the application.

Preprocessing

Image Preprocessing is the next step after a digital image is obtained.

The key function of preprocessing is to improve the image in ways that increase the chances for success of the other processes.

Image Preprocessing typically deals with techniques for enhancing contrast, removing noise, and isolating regions whose texture indicate a likelihood of alphanumeric information.

Example of Preprocessing

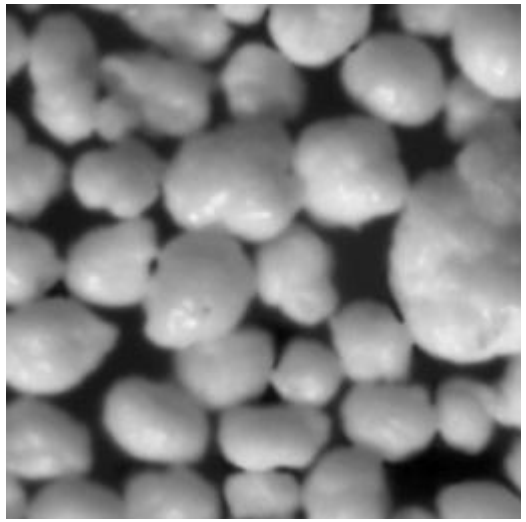
Before Contrast | After Contrast

Segmentation

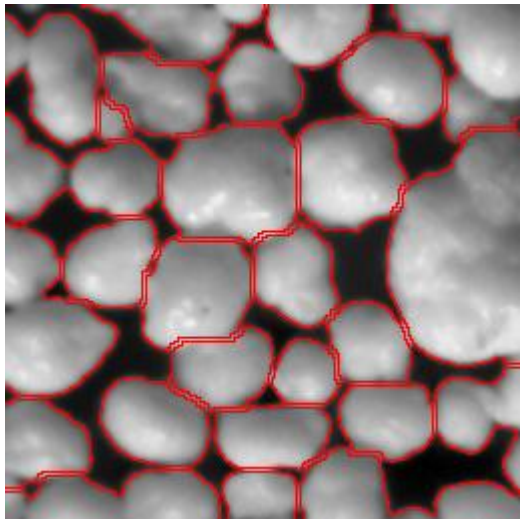
- The next stage deals with ***Segmentation***.
- Broadly defined, segmentation partitions an input image into its constituent parts or objects.
- In general, autonomous segmentation is one of the most difficult tasks in digital image processing.
- A rugged segmentation procedure brings the process a long way towards successful solution of an imaging problem.
- Weak or erratic segmentation algorithms almost always guarantee eventual failure.
- In terms of character recognition, the key role of segmentation is to extract individual characters and words from the background.

Example of Segmentation

Original image



Segmented image



Representation & Description

- The output of the segmentation stage usually is raw pixel data, constituting either the boundary of a region or all the points in the region itself.
- In either case, converting the data to a form suitable for computer processing is necessary.
- The first decision that must be made is whether the data should be represented as a boundary or as a complete region.
- Boundary representation is appropriate when the focus is on external shape characteristics, such as corners and inflections.
- Regional representation is appropriate when the focus is on internal properties, such as texture or skeletal shape.
- In some applications, however, these representations coexist.
- Description, also called feature selection, deals with extracting attributes that result in some quantitative information of interest or are basic for differentiating one class of objects from another.

Recognition & Interpretation

The last stage involves ***Recognition*** and ***Interpretation***.

Recognition is the process that assigns a label to an object based on the information provided by its descriptors.

Interpretation involves assigning meaning to an ensemble of recognized objects.

In terms of our example, identifying a character as, say, a “c” requires associating the descriptors for that character with the label “c”.

Knowledge

Knowledge about a problem domain is coded into an image processing system in the form of a knowledge database.

This knowledge may be as simple as detailing regions of an image where the information of interest is known to be located, thus limiting the search that has to be conducted in seeking that information.

The knowledge base also can be quite complex, such as an interrelated list of all major possible defects in a materials inspection problem or an image database containing high-resolution satellite images of a region in connection with change-detection applications.

In addition to guiding the operation of each processing module, the knowledge base also controls the interaction between modules.

OpenCV (http://opencv.org)

- OpenCV (Open Source Computer Vision Library) is released under a BSD license and hence it’s free for both academic and commercial use.
- It has C++, C, Python and Java interfaces and supports Windows, Linux, Mac OS, iOS and Android.

EmguCV (http://www.emgu.com)

- Emgu CV is a cross platform .Net wrapper to the OpenCV image processing library.
- Allowing OpenCV functions to be called from .NET compatible languages such as C#, VB, VC++, IronPython etc.
- The wrapper can be compiled by Visual Studio, Xamarin Studio and Unity, it can run on Windows, Linux, Mac OS X, iOS, Android and Windows Phone.

Aforge.NET Framework (http://aforgenet.com/)

- AForge.NET is a C# framework designed for developers and researchers in the fields of Computer Vision and Artificial Intelligence.

QUIZ LINK- https://forms.gle/iZxmeT7zRDah9HK69

End of Chapter 1